VMC Vidyamandir Classes SINCE 1986
IIT JEE | MEDICAL | FOUNDATION

CODE A

Review of JEE Main - 3 | [JEE 2022]

Date : 19/10/2020 Maximum Marks : 300

Duration : 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, ⊖ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : ______________________

Roll Number : ______________________

OMR Bar Code Number : ______________________

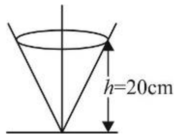
Candidate's Signature : ______________ **Invigilator's Signature** ______________

PART I : PHYSICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A right circular cone is fixed with its axis vertical and vertex down. A particle in contact with its smooth inside surface describes circular motion in a horizontal plane at a height of 20 cm above the vertex. Its velocity in m/s is :



(A) 1 **(B)** 1.2 **(C)** 1.4 **(D)** 1.6

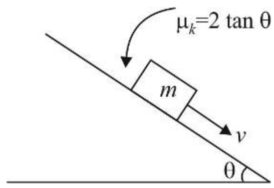
2. A body of mass 2 kg is moving under the influence of a central force whose potential energy is given by $U(r) = 2r^3$ joule. If the body is moving in a circular orbit of radius 5m, then its total energy is:

(A) 625 J **(B)** 225 J **(C)** 325 J **(D)** 525 J

3. A particle of mass *m* is made to move with uniform speed v_0 along the perimeter of a regular hexagon, inscribed in a circle of radius *R*. The magnitude of change in momentum of particle at each corner of the hexagon is:

(A) $2mv_0 \sin\frac{\pi}{6}$ **(B)** $mv_0 \sin\frac{\pi}{6}$ **(C)** $mv_0 \sin\frac{\pi}{3}$ **(D)** $2mv_0 \sin\frac{\pi}{3}$

4. A block is projected with initial velocity v on an incline as shown. Its speed after a time $t = \frac{v}{2g\sin\theta}$ will be:

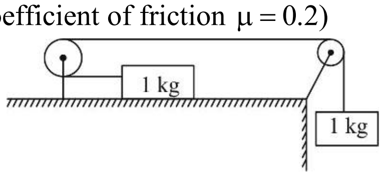


(A) $\frac{v}{3}$ **(B)** $\frac{v}{2}$ **(C)** zero **(D)** $\frac{v}{4}$

5. The bob of a pendulum of length 980 cm is released from rest with its string making an angle 60° with the vertical. At its lowest position, it collides head on with a ball of mass 20 gram resting on a smooth horizontal surface. If the collision is perfectly elastic, with what velocity will the bob move immediately after the collision? Mass of the bob is 10 gram. $(g = 9.8 \text{ m/s}^2)$

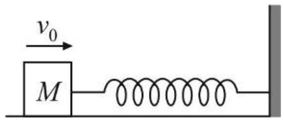
(A) 490 cms^{-1} **(B)** $\frac{2}{3}\times 980 \text{ cms}^{-1}$ **(C)** $\frac{980}{4} \text{ cms}^{-1}$ **(D)** $\frac{980}{3} \text{ cms}^{-1}$

6. A block of mass 1 kg is placed on a rough horizontal surface connected by a light string passing over two smooth pulleys as shown. Another block of 1 kg is connected to the other end of the string. The acceleration of the blocks is: (coefficient of friction $\mu = 0.2$)



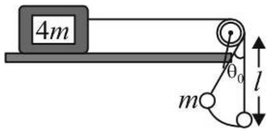
(A) 0.8 g **(B)** 0.4 g **(C)** 0.5 g **(D)** zero

7. 1 kg block collides with a horizontal weightless spring of force constant 100 N/m, as shown in the figure. The block compresses the spring by 0.4 m from the rest position. Assuming that the coefficient of kinetic friction between the block and the horizontal surface is 0.9, the speed of the block at the instant of collision is approximately $(g = 10 \text{ m/s}^2)$:



(A) 4.8 m/s **(B)** 5.6 m/s **(C)** 1.4 m/s **(D)** 5.8 m/s

8. Two bodies of masses m and 4m are attached with string as shown in the figure. The body of mass m hanging from a string of length l is executing oscillations of angular amplitude θ_0, while the other body is at rest. The minimum coefficient of friction between the mass 4m and the horizontal surface should be:

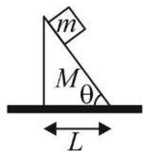


(A) $\left(\dfrac{2-\cos\theta_0}{3}\right)$ **(B)** $2\cos^2\left(\dfrac{\theta_0}{2}\right)$ **(C)** $\left(\dfrac{1-\cos\theta_0}{2}\right)$ **(D)** $\left(\dfrac{3-2\cos\theta_0}{4}\right)$

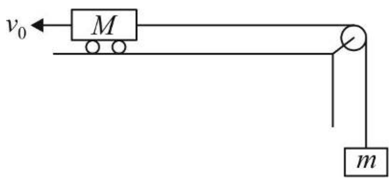
9. A block of mass m is released on a smooth inclined surface of wedge of mass M. Find the displacement of the wedge on horizontal smooth surface when the block reaches the lower end.

(A) $\dfrac{mL}{2(M+m)}$ **(B)** $\dfrac{mL}{M+m}$

(C) $\dfrac{m\sin\theta . L}{2(M+m\cos^2\theta)}$ **(D)** $\dfrac{mL}{2(M+m\cos\theta)}$



10. A cart of mass $M = \dfrac{1}{2}$ kg is connected by a string to a weight of mass $m = \dfrac{1}{5}$ kg. At initial moment, the cart is given a velocity $v_0 = 7$ m/s to the left along the smooth horizontal surface as shown. The magnitude and direction of velocity of cart at $t = 5s$ is: $(g = 9.8 \text{ m/s}^2)$:



(A) 2.8 m/s, rightwards **(B)** 2.8 m/s, leftwards
(C) 7 m/s, rightwards **(D)** 7 m/s, leftwards

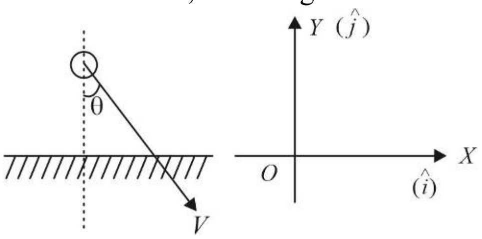
11. A block of mass 2 kg is kept on the floor. The coefficient of static friction is 0.4. If a force F of 2.5 N is applied on the block as shown in the figure, the frictional force between the block and the floor will be:

(A) 2.5 N **(B)** 5 N **(C)** 7.84 N **(D)** 10 N

12. A body of mass 10 kg is moving along a circular path of radius 100 m. Its speed increases, in a uniform manner, from 17 m/s to 26 m/s in a time duration of 3s. Force acting on the body when it is travelling at a speed 20 m/s is:

(A) 45 newton **(B)** 42.5 newton **(C)** 47.5 newton **(D)** 50 newton

13. A smooth ball strikes obliquely with a velocity v_0 at an angle θ with vertical with a horizontal surface. If 'e' is coefficient of restitution of collision, the change in momentum of ball is:



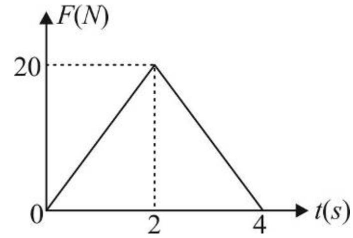
(A) $mv_0 \cos\theta(1+e)\hat{j}$ **(B)** $mv_0 \sin\theta(1-e)\hat{j}$

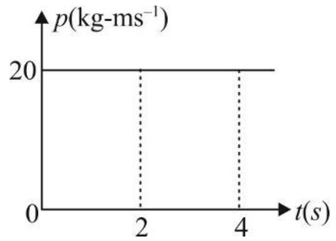
(C) $-mv_0(1+e)\sin\theta\,\hat{j}$ **(D)** None of these

14. The momentum of a particle is given by $\vec{P} = (4\sin t\,\hat{i} - 4\cos t\,\hat{j})$ kg m/s. Select the correct alternative:

(A) Momentum P of the particle is always parallel to F.
(B) Momentum P of the particle is always perpendicular to F.
(C) Magnitude of momentum P is variable.
(D) None of the above

15. The graph shows the variation of force acting on a body with time t. Assuming body to start from rest, the variation of its momentum with time is best represented by :



(A)


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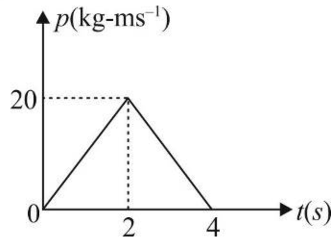
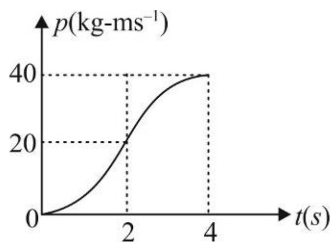
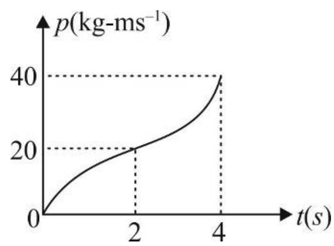
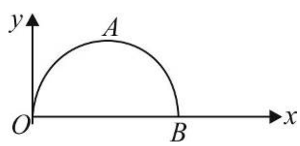


(C)


(D)


16. Given $\vec{F} = (xy^2)\hat{i} + (x^2 y)\hat{j}$ newton.

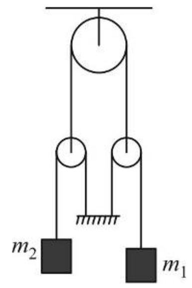
The work done by $\vec{F}$ when a particle is taken along the semi-circular path OAB where the co-ordinates of B are $(4m, 0m)$ is:



(A) $\frac{65}{3}J$ **(B)** $\frac{75}{2}J$

(C) $\frac{73}{4}J$ **(D)** zero

17. In the given figure, all strings and pulleys are ideal and acceleration of m_1 is $\frac{g}{3}$ m/s^2 upward. Then find the ratio of m_1 / m_2.



(A) $\frac{1}{3}$ **(B)** 1

(C) $\frac{1}{2}$ **(D)** $\frac{1}{4}$

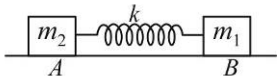
18. A body of mass m is projected vertically up with a speed v_0. If it passes through the point of projection with a speed $\frac{v_0}{2}$, the work done by air resistance is :

(A) zero **(B)** $\frac{-3mv_0^2}{4}$ **(C)** $\frac{3mv_0^2}{4}$ **(D)** $\frac{-3mv_0^2}{8}$

19. The potential energy of particle of mass 0.1 kg moving along the x-axis is given by $U = 5x(x-4)J$, where x is in metre. It can be concluded that :

(A) the particle is acted upon by a constant force
(B) the speed of the particle is maximum at $x = 2m$
(C) the particle is in stable equilibrium at $x = 3$
(D) the particle is in unstable equilibrium at $x = 4$

20. Two bodies (at rest) A and B of masses m_1 and m_2 respectively are connected by a massless spring of force constant k over a smooth surface. A constant force F starts acting on the body A at $t = 0$. Then :



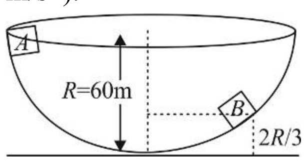
(A) at every instant, the acceleration of centre of mass is $\frac{F}{m_1 + m_2}$
(B) just after the application of force, accelerations of A and B is zero
(C) the acceleration of A is constant
(D) the acceleration of B is constant

SECTION-2

This section has Five (05) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*.
(Example: 6, 81, 1.50, 3.25, 0.08)

1. A motor pump is used to deliver water at a certain rate from a given pipe. To obtain 5 times water from the same pipe in the same time, the power of motor must be increased to __________ times.

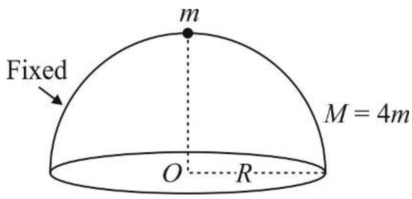
2. A particle of mass m is released from rest at point A along the inside surface of a fixed smooth hemispherical bowl of radius R = 60 m. The speed at B which is at a height $h = \frac{2R}{3}$ from the lowest point is _______ m/s. (Take $g = 10$ m/s^2).



3. A metal ball of mass 2 kg moving with a velocity of 36 km/h has a head on collision with a stationary ball of mass 3 kg. If after the collision, the two balls move together, the loss in kinetic energy due to collision is ________ (in joule).

4. A ball strikes a frictionless horizontal floor at an angle $\theta = 45°$. The coefficient of restitution between the ball and the floor is $e = \frac{1}{2}$. The percentage of its kinetic energy lost in collision is ________.

5. A small particle of mass m starts sliding down from rest at top along the smooth surface of a fixed hollow hemisphere of mass M (= $4m$). The distance of centre of mass of (particle + hemisphere) from centre O of base of hemisphere, when the particle separates from the surface of hemisphere is $\frac{\sqrt{69}}{5\alpha}R$. Find the value of α. The CM of a hollow hemisphere lies at a distance $R/2$ from centre of base on the symmetry axis.



PART II : CHEMISTRY 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Which molecule contains both ionic and covalent bond?

(A) Al_4C_3 **(B)** CaF_2 **(C)** CaH_2 **(D)** CaC_2

2. The main axis of diatomic molecule which connect the nucleus of two atoms is y, the orbital p_z and p_z overlap to form which of the following orbital?

(A) π molecular orbital **(B)** σ molecular orbital
(C) δ molecular orbital **(D)** No bond will form

3. In which of the following pair both the species have sp^3 hybridization?

(A) HCN, NO_3^- **(B)** BF_4^-, $HgCl_2$ **(C)** ClO_4^-, OF_2 **(D)** NO_2, SO_2

4. Which of the following is correct order of size?

(A) $Li^+(g) < H^-(g)$ (B) $Li^+(g) < Na^+(g)$

(C) $Li^+(aq) > Na^+(aq)$ (D) $F^-(aq) > Cl^-(aq)$

(A) Only (C) **(B)** Only (A) **(C)** Only (B), (C) **(D)** (A), (B), (C), (D)

5. The correct order in which the bond order increases in the following is

(A) $O_2^{2-} < O_2 < O_3$ **(B)** $O_3 < O_2^{2-} < O_2$ **(C)** $O_2^{2-} < O_3 < O_2$ **(D)** $O_2 < O_2^{2-} < O_3$

6. Which of the following would have a permanent dipole moment?

(A) CS_2 **(B)** $TeCl_4$ **(C)** SF_6 **(D)** IF_7

7. The formal charge of sulphur in H_2SO_4 and phosphorus in H_3PO_4 respectively are:

(A) – 1, +1 **(B)** 0, 0 **(C)** +2, +1 **(D)** 0, +1

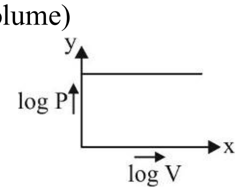
8. The density of ideal gas will be highest at :

(A) 273 K and 1 atm **(B)** 273 K and 2 atm
(C) 273°C and 1 atm **(D)** 273°C and 2 atm

9. Oxygen gas is sixteen times heavier than a hydrogen gas. At room temperature, the average kinetic energy of oxygen gas is :

(A) two times that of a hydrogen gas **(B)** same as that of a hydrogen gas
(C) four times that of a hydrogen gas **(D)** half that of a hydrogen gas

10. For 1 mol of an ideal gas at a constant temperature T, the plot of (log P) against (log V) is a (P: Pressure, V : Volume)

(A)


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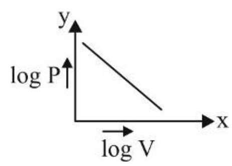

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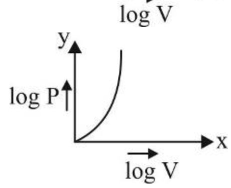

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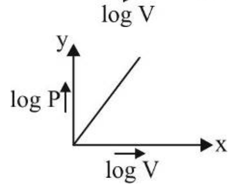


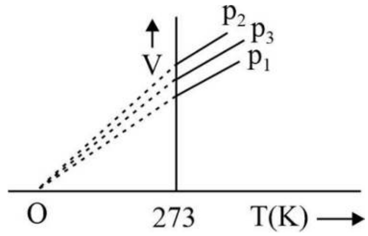
11. The ratio between the root mean square speed and average speed of O_2 at 800 K is:

(A) $\left(\frac{8}{3}\right)^{1/2}$ **(B)** $\left(\frac{8}{3\pi}\right)^{1/2}$ **(C)** $\left(\frac{3\pi}{8}\right)^{1/2}$ **(D)** $\left(\frac{8\pi}{3}\right)^{1/2}$

12. A bubble of air rises to the surface where the temperature is 298K and the pressure is 1.0 bar. What happened to the volume of the bubble when its initial underwater temperature was 288 K & pressure 1.5 bar ?

(A) Volume of bubble decreased by half of the initial volume
(B) Volume of bubble remained same
(C) Volume of bubble increased by factor 1.6
(D) Volume of bubble decreased by factor 1.6

13. The volume-temperature graphs of a given mass of an ideal gas at constant pressure are shown below.
What is the correct order of pressures?



(A) $p_1 > p_3 > p_2$ **(B)** $p_1 > p_2 > p_3$
(C) $p_2 > p_3 > p_1$ **(D)** $p_2 > p_1 > p_3$

14. "One gram atom of a helium gas at S.T.P. occupies 22.4 litres". This fact was derived from

(A) Dalton's theory **(B)** Avogadro's hypothesis
(C) Berzelius hypothesis **(D)** Law of gaseous volume

15. Equal masses of methane and oxygen are mixed in an empty container at 25°C, the pressure exerted by oxygen is $\frac{1}{3}$ rd of total pressure. Find out the mole fraction of oxygen in container.

(A) 1/2 **(B)** 2/3 **(C)** 1/5 **(D)** 1/3

16. $d\pi - p\pi$ bond present in

(A) SO_4^{2-} **(B)** PO_4^{3-} **(C)** SO_2 **(D)** All of these

17. How many gram atoms of Fe in 100g Haemoglobin if it contains 0.33% Fe. (atomic mass of Fe = 56)

(A) 5.89×10^{-3} **(B)** 0.035×10^{23} **(C)** 3.5×10^{23} **(D)** 7×10^{8}

18. The hybrid orbital having only 60% p character in :

(A) C_6H_6 **(B)** XeO_2F_2 **(C)** ClO_3^- **(D)** HCHO

19. How many mL of 0.5 N HCl acid will be required to completely neutralise 500ml of a 0.1N NaOH solution?

(A) 50 ml **(B)** 100 ml **(C)** 150 ml **(D)** 200 ml

20. What is the dominant intermolecular force or bond that must be responsible for ice being less dense than water?

(A) Dipole-dipole interaction **(B)** Covalent bonds
(C) London force **(D)** Hydrogen bonding

SECTION-2

This section has Five (05) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

1. In the following reaction: $MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$. When 2 moles of MnO_2 reacted with 4 moles of HCl, 11.2L Cl_2 was collected at STP. Find the percent yield of Cl_2.

2. A compound H_2X with molar weight of 60 g is dissolved in a solvent having density of $0.8 \, g \, mL^{-1}$. Assuming no change in volume upon dissolution, the molality of a 4.8 molar solution is :

3. Among NO, CN^-, H_2, He_2^+, Li_2, Be_2, B_2, C_2, N_2, O_2^-, NO^+, O_2^{2-} and F_2, the number of diamagnetic species is :

4. 50 % pure sample of $CaCO_3$ heated with 80 % yield to produce 0.448 litre CO_2 at 1 atm and 273 K temperature. Calculate mass of $CaCO_3$ sample in gm.

5. Identify the number of amphiprotic species in the following list?

H_3O^+, HPO_4^{2-}, HCO_3^-, H_2O, HPO_3^{2-}, $H_2PO_2^-$, $H_2PO_4^-$, $H_2PO_3^-$, PO_4^{3-}, NH_4^+

PART III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Find the rank of the word 'CROMA' if arranged in dictionary order.

(A) 48 **(B)** 57 **(C)** 39 **(D)** None of these

2. $\sum_{r=1}^{180} \cos(r^\circ) =$

(A) 0 **(B)** 1 **(C)** -1 **(D)** None of these

3. Find the no. of Non-Negative integral solutions of the inequality:

$x + y + z \leq 3$

(A) 15 **(B)** 20 **(C)** 25 **(D)** None of these

4. If $P_n = \cos^n \theta + \sin^n \theta$, then $P_n - P_{n-2} = kP_{n-4}$, where k is equal to:

(A) 1 **(B)** $-\sin^2 \theta \cos^2 \theta$ **(C)** $\sin^2 \theta$ **(D)** $\cos^2 \theta$

5. No. of diagonals in a regular octagon is m then no. of divisors of m is:

(A) 5 **(B)** 8 **(C)** 6 **(D)** 9

6. There are 10 students in a class in which three A, B, C are girls. The number of ways to arrange them in a row when no two girls together:

(A) $7! \times {}^8P_3$ **(B)** $7! \times {}^3P_3$ **(C)** $10! \times {}^3P_3$ **(D)** None of these

7. If ${}^{n^2-n}C_2 = {}^{n^2-n}C_{10}$, then n equals:

(A) 12 **(B)** 4 only **(C)** -3 only **(D)** $4 \text{ or } -3$

8. No. of square in a CHESS board is n then biggest prime divisor of n is:

(A) 11 **(B)** 13 **(C)** 17 **(D)** 19

9. The number of 4 digits odd numbers formed with the digits 0,1,2,3, 4 and 5 is: (Repetition not allowed)

(A) 54 **(B)** 144 **(C)** 180 **(D)** 360

10. In how many ways 7 different beads be strung into a ring so that two particular beads are always together?

(A) 240 **(B)** 720 **(C)** 120 **(D)** 360

11. The number of words which can be formed from the letters of the word 'JODHPUR' so that P, U, R always remain together is:

(A) 4320 **(B)** 120 **(C)** 720 **(D)** None of these

12. The number of words formed with the letters of the word 'JODHPUR' in which the three letters P, U, R never come all together, is:

(A) 720 **(B)** 5040 **(C)** 4320 **(D)** None of these

13. The number of ways in which six different prizes can be distributed among three children each receiving at least one prize is:

(A) 270 **(B)** 540 **(C)** 729 **(D)** 539

14. A quadratic equation whose roots are $\operatorname{cosec}^2\alpha$ and $\sec^2\beta$, can be:

(A) $x^2 - 2x + 2 = 0$ **(B)** $x^2 - 3x + 3 = 0$

(C) $x^2 - 5x + 5 = 0$ **(D)** $x^2 + 4x - 4 = 0$

15. If $a = \sin 170° + \cos 170°$, then:

(A) $a > 0$ **(B)** $a < 0$ **(C)** $a = 0$ **(D)** $a = 1$

16. If $A = \tan 6° \tan 42°$ and $B = \cot 66° \cot 78°$ then:

(A) $A = 2B$ **(B)** $A = \frac{1}{3}B$ **(C)** $A = B$ **(D)** $3A = 2B$

17. $\cos 52° + \cos 68° + \cos 172° =$

(A) 0 **(B)** 1 **(C)** 2 **(D)** None of these

18. The number of real solutions of the equation $\sin\left(e^x\right) = 2^x + 2^{-x}$ is:

(A) 1 **(B)** 0 **(C)** 2 **(D)** Infinite

19. The value of $\tan\frac{\pi}{3} + 2\tan\frac{2\pi}{3} + 4\tan\frac{4\pi}{3} + 8\tan\frac{8\pi}{3}$ is equal to:

(A) $-5\sqrt{3}$ **(B)** $\frac{-5}{\sqrt{3}}$ **(C)** $5\sqrt{3}$ **(D)** $\frac{5}{\sqrt{3}}$

20. If no. of zeroes at the end of $75!$ is n then value of $\cos(2n)° - \sin n° =$

(A) 0 **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{2}$ **(D)** $-\frac{1}{4}$

SECTION-2

This section has Five (05) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****.**
(Example: 6, 81, 1.50, 3.25, 0.08)

1. The number of perfect square divisors of 9600 is:

2. There are 12 points in a plane. The number of the straight lines joining any two of them when 3 of them are collinear is:

3. The number of permutations formed without changing the positions of vowel and consonants of the letter of word ‘ALGEBRA’ is:

4. There are four balls of different colours and four boxes of colours same as those of the balls. The number of ways in which the balls, one in each box, could be placed such that a ball does not go to box of its own colour is:

5. The number of integral values of ‘n’ so that $\sin x (\sin x + \cos x) = n$ has at least one solution, is:

End of Review of JEE Main - 3 | JEE 2022